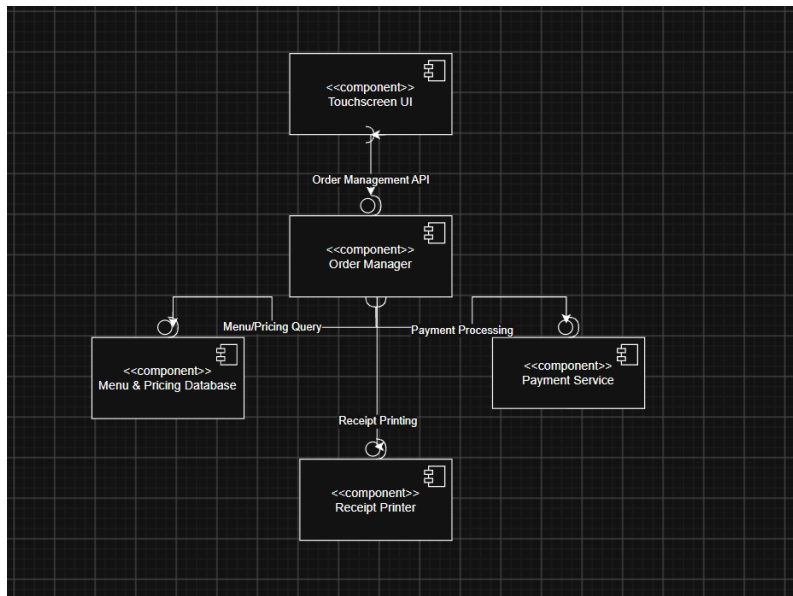


LAB 3 — COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Self-Service Coffee Kiosk System



Architecture Selection

Selected Architecture: Layered Architecture

The Self-Service Coffee Kiosk is organized into clearly separated responsibilities: customer interaction, order management, payment processing, menu and pricing data, and receipt printing.

Why Layered Architecture?

- 1 **Separation of concerns:** Each component has a focused responsibility, making the system easier to understand, maintain, and modify.
- 2 **Suitability:** The kiosk has focused functionality—drink selection, size selection, price calculation, payment, and receipt printing—so a straightforward layered structure is appropriate.

Security Advantage

Payment processing is isolated in the **Payment Service** component. The Order Manager communicates through a defined interface, keeping payment-processing implementation separate from the customer-facing interface.

Performance Benefit

Ordering, payment, database access, and receipt printing are handled by dedicated components. This separation keeps responsibilities focused and supports responsive touchscreen interactions.

Components & Interfaces

Components: Touchscreen UI • Order Manager • Payment Service • Menu & Pricing Database • Receipt Printer

Interfaces: Order Management API • Payment Processing • Menu/Pricing Query • Receipt Printing

Deliverable: UML component diagram with provided/required interfaces and component dependencies.